

Adaptive Resonance Theory

Stability/Plasticity

- A key problem of the Grossberg networks is that they do **not always form stable clusters** (or categories).
- Grossberg did show that if the number of input patterns is not too large, or if the input patterns do not form too many clusters relative to the number of neurons in Layer 2, then the learning eventually stabilizes.
- However, he also showed that the standard competitive networks **do not have stable learning** in response to arbitrary (**based on random choice**) input patterns.
- The learning instability occurs because **of the network's adaptability (or plasticity)**, which causes prior learning to be eroded by more recent learning.
- Grossberg refers to this problem as the “***stability/plasticity dilemma***.”

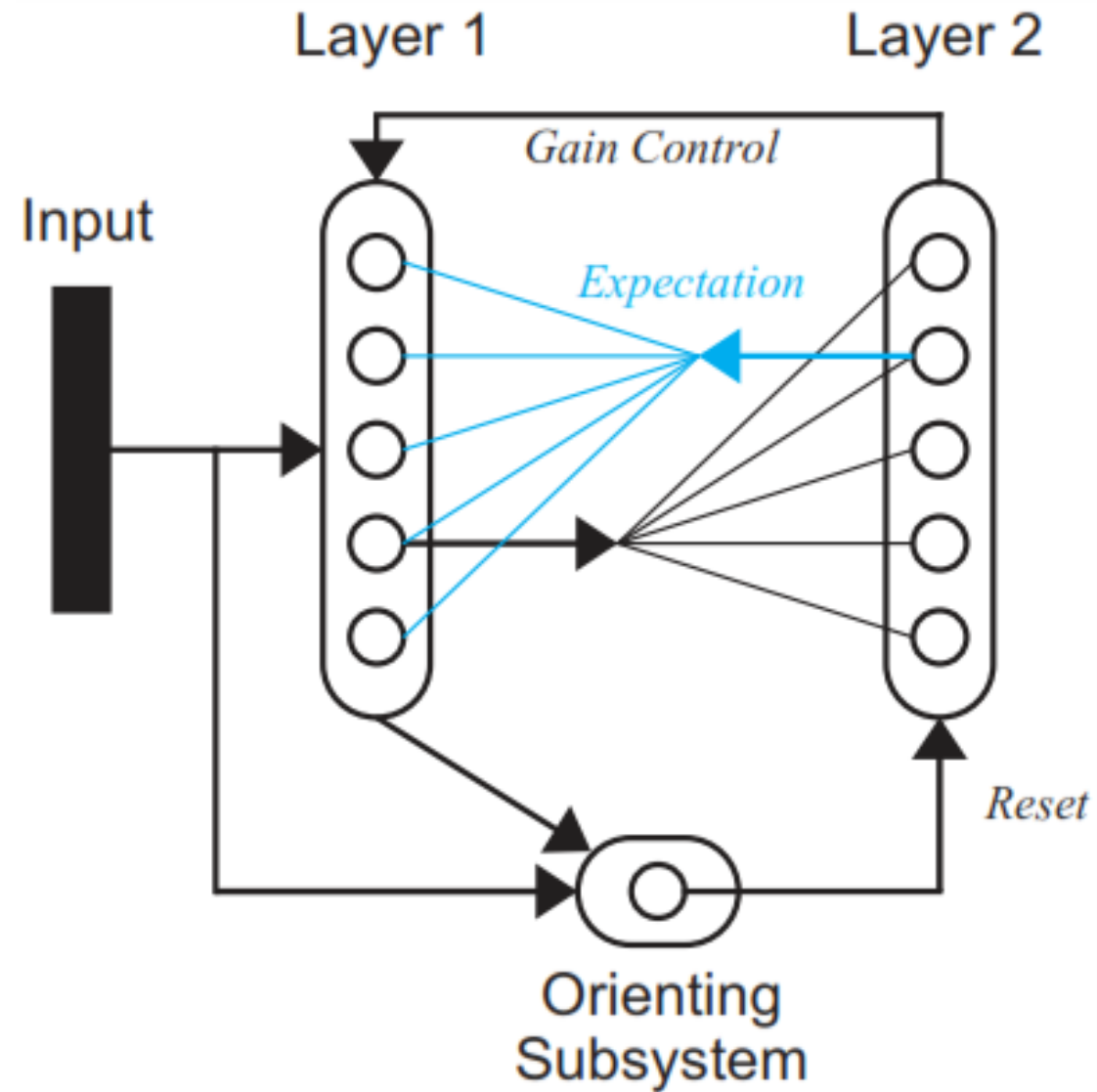
Adaptive Resonance Theory (ART)

- ART developed by Grossberg and Gail Carpenter developed to address the stability/plasticity dilemma
- ART networks are based on the Grossberg network
- ART key innovation is the use of “expectations”.
 - As each input pattern is presented to the network, it is compared with the prototype vector that it most closely matches (the expectation).
 - If the match between the prototype and the input vector is not adequate, a new prototype is selected.
- In this way, previously learned memories (prototypes) are not eroded by new learning.

Overview of Adaptive Resonance

- The basic ART architecture is a modification of the Grossberg network
- ART is designed to stabilize the learning process.
- The innovations of the ART architecture consist of three parts:
 - Layer 2 (L2) to Layer 1 (L1) expectations,
 - The orienting subsystem
 - Gain control.

Basic ART Architecture



General operation of the ART system

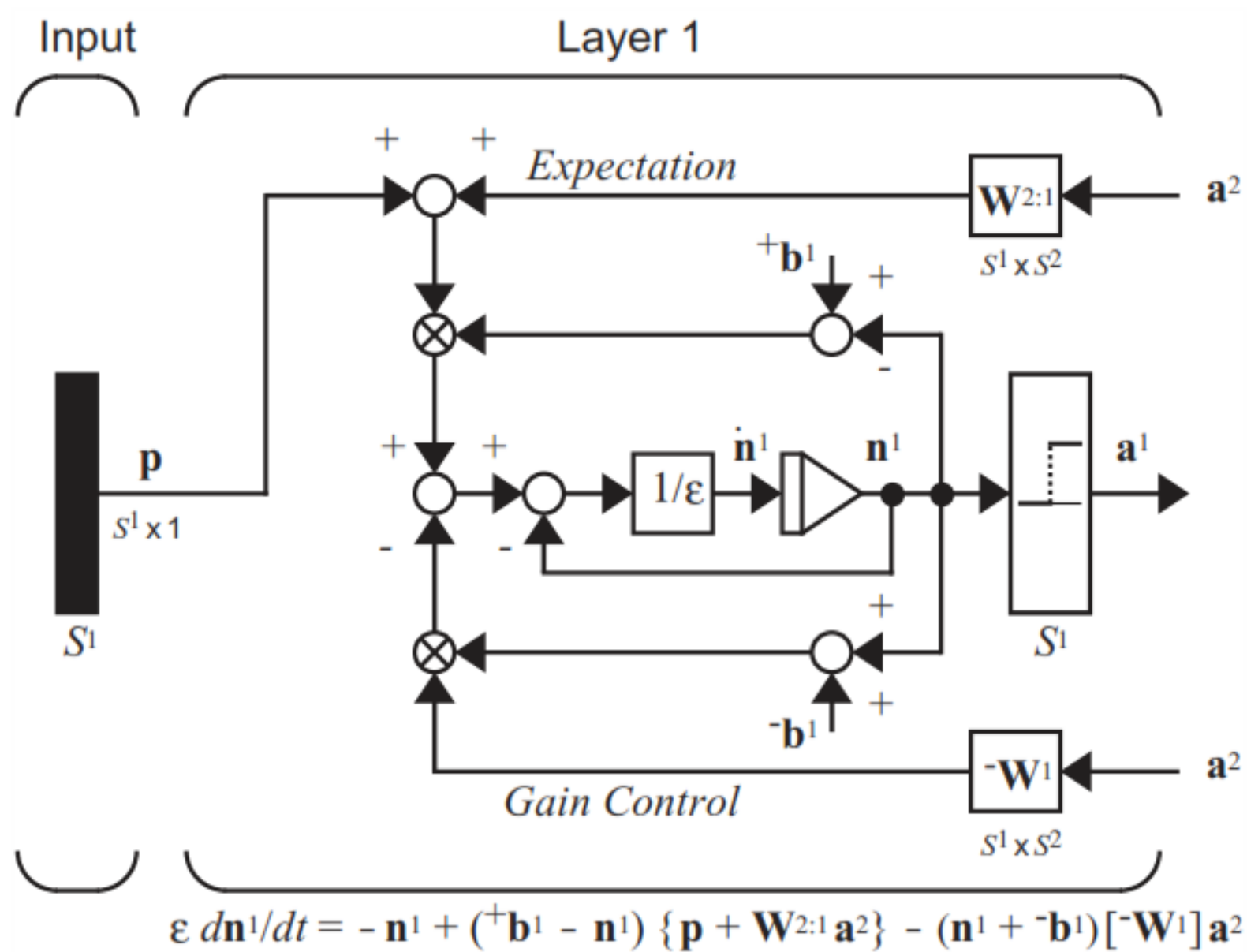
- The L1-L2 connections of the Grossberg network are instars (Hint: have a vector input and a scalar output).
- They perform a clustering (or categorization) operation.
 1. When an input pattern is presented to the network, it is multiplied (after normalization) by the L1-L2 weight matrix.
 2. Then, a competition is performed at Layer 2 to determine which row of the weight matrix is closest to the input vector.
 3. That row is then moved toward the input vector.
- After learning is complete, each row of the L1-L2 weight matrix is a prototype pattern, which represents a cluster (or category) of input vectors.
- In the ART networks, learning also occurs in a set of feedback connections from Layer 2 to Layer 1.

- These connections are outstars, which perform pattern recall.
(Hint- outstars: have a scalar input and a vector output)
1. When a node in Layer 2 is activated, this reproduces a prototype pattern (the expectation) at Layer 1.
 2. Layer 1 then performs a comparison between the expectation and the input pattern.
 3. When the expectation and the input pattern are **not closely matched**, the **orienting subsystem** causes a **reset in Layer 2**.
 4. This reset disables the current winning neuron, and the current expectation is removed.
 5. A new competition is then performed in Layer 2, while the previous winning neuron is disabled.
 6. The new winning neuron in Layer 2 projects a new expectation to Layer 1, through the L2-L1 connections.
 7. This process continues until the L2-L1 expectation provides a close enough match to the input pattern.

ART Subsystems

- **Layer 1**
 - Normalization
 - Comparison of input pattern and expectation
- **L1-L2 Connections (Instars)**
 - Perform clustering operation
 - Each row of $W^{1:2}$ is a prototype pattern
- **Layer 2**
 - Competition, contrast enhancement
- **L2-L1 Connections (Outstars)**
 - Expectation
 - Perform Pattern Recall
 - Each column of $W^{2,1}$ is a prototype pattern
- **Orienting Subsystem**
 - Causes a reset when expectation does not match input
 - Disables current winning neuron

ART1 – Layer 1



Purpose of Layer 1

- The main purpose of Layer 1 is to compare the input pattern with the expectation pattern from Layer 2. (*Both patterns are binary in ART1*)
- If the patterns are not closely matched, the **orienting subsystem will cause a reset** in Layer 2.
- If the patterns are close enough, Layer 1 combines the expectation and the input to form a new prototype pattern.

Layer 1 operation

Shunting Model

$$\varepsilon \frac{d\mathbf{n}^1(t)}{dt} = -\mathbf{n}^1(t) + (\mathbf{b}^1 - \mathbf{n}^1(t)) \underbrace{\{\mathbf{p} + \mathbf{W}^{2:1} \mathbf{a}^2(t)\}}_{\substack{\text{Excitatory Input} \\ \text{(Comparison with Expectation)}}} - (\mathbf{n}^1(t) + \mathbf{b}^1) \underbrace{[\mathbf{W}^1] \mathbf{a}^2(t)}_{\substack{\text{Inhibitory Input} \\ \text{(Gain Control)}}}$$

$$\mathbf{a}^1 = \mathbf{hardlim}^+(\mathbf{n}^1)$$

$$\mathbf{hardlim}^+(n) = \begin{cases} 1, & n > 0 \\ 0, & n \leq 0 \end{cases}$$

Excitatory input to Layer 1

$$\mathbf{p} + \mathbf{W}^{2:1} \mathbf{a}^2(t)$$

Suppose that neuron j in Layer 2 has won the competition:

$$\mathbf{W}^{2:1} \mathbf{a}^2 = \begin{bmatrix} \mathbf{w}_1^{2:1} & \mathbf{w}_2^{2:1} & \dots & \mathbf{w}_j^{2:1} & \dots & \mathbf{w}_{S^2}^{2:1} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \end{bmatrix} = \mathbf{w}_j^{2:1} \quad (j\text{th column of } \mathbf{W}^{2:1})$$

Therefore the excitatory input is the sum of the input pattern and the L2-L1 expectation:

$$\mathbf{p} + \mathbf{W}^{2:1} \mathbf{a}^2 = \mathbf{p} + \mathbf{w}_j^{2:1}$$

Each column of the L2-L1 matrix represents a different expectation (prototype pattern)

Layer 1 combines the input pattern with the expectation using AND operation

Inhibitory input to Layer 1

Gain Control

$$[\tilde{\mathbf{W}}^1] \mathbf{a}^2(t)$$

$$\tilde{\mathbf{W}}^1 = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}$$

The gain control will be one when Layer 2 is active (one neuron has won the competition), and zero when Layer 2 is inactive (all neurons having zero output).

Steady State Analysis

$$\varepsilon \frac{d\mathbf{n}^1(t)}{dt} = -\mathbf{n}^1(t) + (\mathbf{b}^1 - \mathbf{n}^1(t))\{\mathbf{p} + \mathbf{W}^{2:1}\mathbf{a}^2(t)\} - (\mathbf{n}^1(t) + \mathbf{b}^1)[\mathbf{W}^1]\mathbf{a}^2(t)$$

- The response of neuron i in Layer 1 is described by

$$\varepsilon \frac{dn_i^1}{dt} = -n_i^1 + (b^1 - n_i^1) \left\{ p_i + \sum_{j=1}^{s^2} w_{i,j}^{2:1} a_j^2 \right\} - (n_i^1 + b^1) \sum_{j=1}^{s^2} a_j^2$$

- The steady state response of this system for **two different cases** are:
 - 1st. Case: Layer 2 is inactive, therefore $a_j^2 = 0$ for all j .
 - 2nd. Case: Layer 2 is active, and therefore one neuron has an output of 1, and all other neurons output 0.

1st. Case, Layer 2 is inactive: therefore $a_j^2 = 0$ for all j

$$\varepsilon \frac{dn_i^1}{dt} = -n_i^1 + ({}^+b^1 - n_i^1)\{p_i\}.$$

In the steady state ($dn_i^1(t)/dt = 0$) we have

$$0 = -n_i^1 + ({}^+b^1 - n_i^1)p_i = -(1 + p_i)n_i^1 + {}^+b^1 p_i.$$

If we solve for the steady state neuron output n_i^1 we find

$$n_i^1 = \frac{{}^+b^1 p_i}{1 + p_i}.$$

Therefore, if $p_i = 0$ then $n_i^1 = 0$, and if $p_i = 1$ then $n_i^1 = {}^+b^1/2 > 0$. Since we chose the transfer function for Layer 1 to be the *hardlim⁺* function, then we have

$$\mathbf{a}^1 = \mathbf{p}.$$

Therefore, when Layer 2 is inactive, the output of Layer 1 is the same as the input pattern.

2nd. Case, Layer 2 is active: neuron j is the winner in Layer 2,
therefore $a_j^2 = 1$ and $a_k^2 = 0$ for all $k \neq j$

$$\varepsilon \frac{dn_i^1}{dt} = -n_i^1 + ({}^+b^1 - n_i^1)\{p_i + w_{i,j}^{2:1}\} - (n_i^1 + {}^-b^1)$$

In the steady state ($dn_i^1(t)/dt = 0$) we have

$$\begin{aligned} 0 &= -n_i^1 + ({}^+b^1 - n_i^1)\{p_i + w_{i,j}^{2:1}\} - (n_i^1 + {}^-b^1) \\ &= -(1 + p_i + w_{i,j}^{2:1} + 1)n_i^1 + ({}^+b^1(p_i + w_{i,j}^{2:1}) - {}^-b^1) \end{aligned}$$

If we solve for the steady state neuron output n_i^1 we find

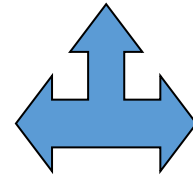
$$n_i^1 = \frac{{}^+b^1(p_i + w_{i,j}^{2:1}) - {}^-b^1}{2 + p_i + w_{i,j}^{2:1}}.$$

- Recall that Layer 1 should combine the input vector with the expectation from Layer 2 (represented by $w_j^{2:1}$).
- Since we are dealing with binary patterns (both the input and the expectation), we will use a logical AND operation to combine the two vectors.
- In other words, we want:
 - n_i^1 to be less than zero when either p_i or $w_{i,j}^{2:1}$ is equal to zero,
 - n_i^1 to be greater than zero when both p_i and $w_{i,j}^{2:1}$ are equal to one.

$$n_i^1 = \frac{{}^+b^1(p_i + w_{i,j}^{2:1}) - {}^-b^1}{2 + p_i + w_{i,j}^{2:1}}$$

$$P_i = 0 \quad \text{or} \quad w_{i,j}^{2:1} = 0$$

$${}^+b^1 - {}^-b^1 < 0$$



$$P_i = 1 \quad \& \quad w_{i,j}^{2:1} = 1$$

$${}^+b^1(2) - {}^-b^1 > 0$$

This can be combine to produce

$${}^+b^1(2) > {}^-b^1 > {}^+b^1$$

Therefore, if this is satisfied, (example: ${}^+b^1=1$ and ${}^-b^1=1.5$) and neuron j of Layer 2 is active, then the output of Layer 1 will be

$$\mathbf{a}^1 = \mathbf{p} \cap \mathbf{w}_j^{2:1}$$

where $\cap$ represents the logical AND operation.

Layer Summery

If Layer 2 is inactive (each $a_j^2 = 0$)

$$\mathbf{a}^1 = \mathbf{p}$$

If Layer 2 is active (one $a_j^2 = 1$)

$$\mathbf{a}^1 = \mathbf{p} \cap \mathbf{w}_j^{2:1}$$

Layer 1 Example:

$$\varepsilon \frac{dn_i^1}{dt} = -n_i^1 + ({}^+b^1 - n_i^1) \left\{ p_i + \sum_{j=1}^{s^2} w_{i,j}^{2:1} a_j^2 \right\} - (n_i^1 + {}^-b^1) \sum_{j=1}^{s^2} a_j^2$$

$\varepsilon = 1, {}^+b^1 = 1 \text{ and } {}^-b^1 = 1.5$
 $\mathbf{w}^{2:1} = \begin{bmatrix} 1 & \mathbf{1} \\ 0 & \mathbf{1} \end{bmatrix}$
 $\mathbf{p} = \begin{bmatrix} \mathbf{0} \\ \mathbf{1} \end{bmatrix} \begin{matrix} \leftarrow p_1 \\ \leftarrow p_2 \end{matrix}$

$j = 2$
 $\downarrow$

Assume that Layer 2 is active, and neuron 2 won the competition.

$$(0.1) \frac{dn_1^1}{dt} = -n_1^1 + (1 - n_1^1) \{ p_1 + w_{1,2}^{2:1} \} - (n_1^1 + 1.5)$$

$$= -n_1^1 + (1 - n_1^1) \{ \mathbf{0} + 1 \} - (n_1^1 + 1.5) = -3n_1^1 - 0.5$$

$$\left. \vphantom{\frac{dn_1^1}{dt}} \right\} \frac{dn_1^1}{dt} = -30n_1^1 - 5$$

$$(0.1) \frac{dn_2^1}{dt} = -n_2^1 + (1 - n_2^1) \{ p_2 + w_{2,2}^{2:1} \} - (n_2^1 + 1.5)$$

$$= -n_2^1 + (1 - n_2^1) \{ \mathbf{1} + 1 \} - (n_2^1 + 1.5) = -4n_2^1 + 0.5$$

$$\left. \vphantom{\frac{dn_2^1}{dt}} \right\} \frac{dn_2^1}{dt} = -40n_2^1 + 5$$

Response of Layer 1

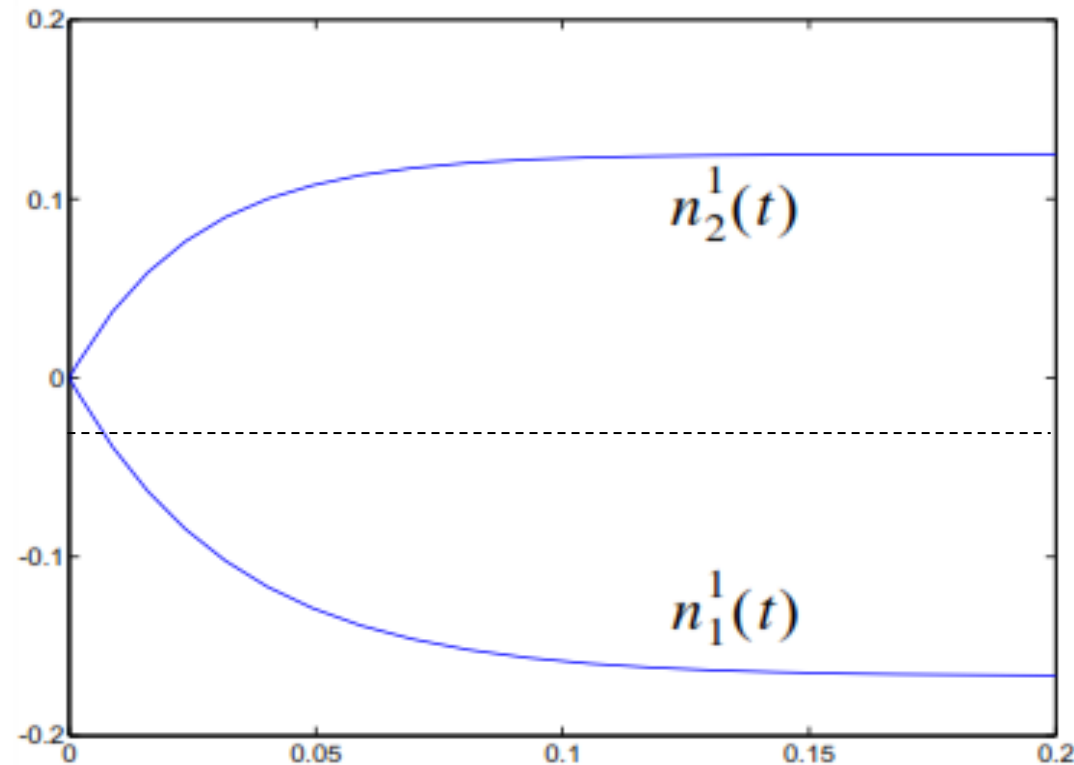
- If we assume that both neurons start with zero initial conditions, the solutions are

$$n_1^1(t) = e^{-30t} n(0) + \int_0^t e^{-30(t-\tau)} (-5) d\tau \quad n_1^1(0) = 0$$

$$n_1^1(t) = \frac{-5 * 6}{6} \int_0^t e^{-30(t-\tau)} d\tau = -\frac{1}{6} (e^{-30(t-t)} - e^{-30(t-0)}) = -\frac{1}{6} (1 - e^{-30t})$$

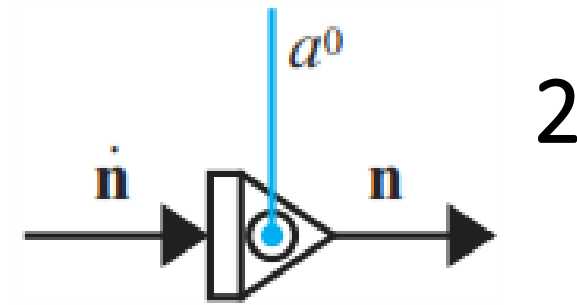
$$n_2^1(t) = e^{-40t} n(0) + \int_0^t e^{-40(t-\tau)} (5) d\tau \quad n_2^1(0) = 0$$

$$n_2^1(t) = \frac{5 * 8}{8} \int_0^t e^{-40(t-\tau)} d\tau = \frac{1}{8} (e^{-40(t-t)} - e^{-40(t-0)}) = \frac{1}{8} (1 - e^{-40t})$$



Note that $n_1^1(t)$ converges to a negative value, and $n_2^1(t)$ converges to a positive value. Therefore, $a_1^1(t)$ converges to 0, and $a_2^1(t)$ converges to 1 (recall that the transfer function for Layer 1 is hardlim^+). This agrees with our steady state analysis $\text{hardlim}^+([0, 1]^T) = [0, 1]^T$ since

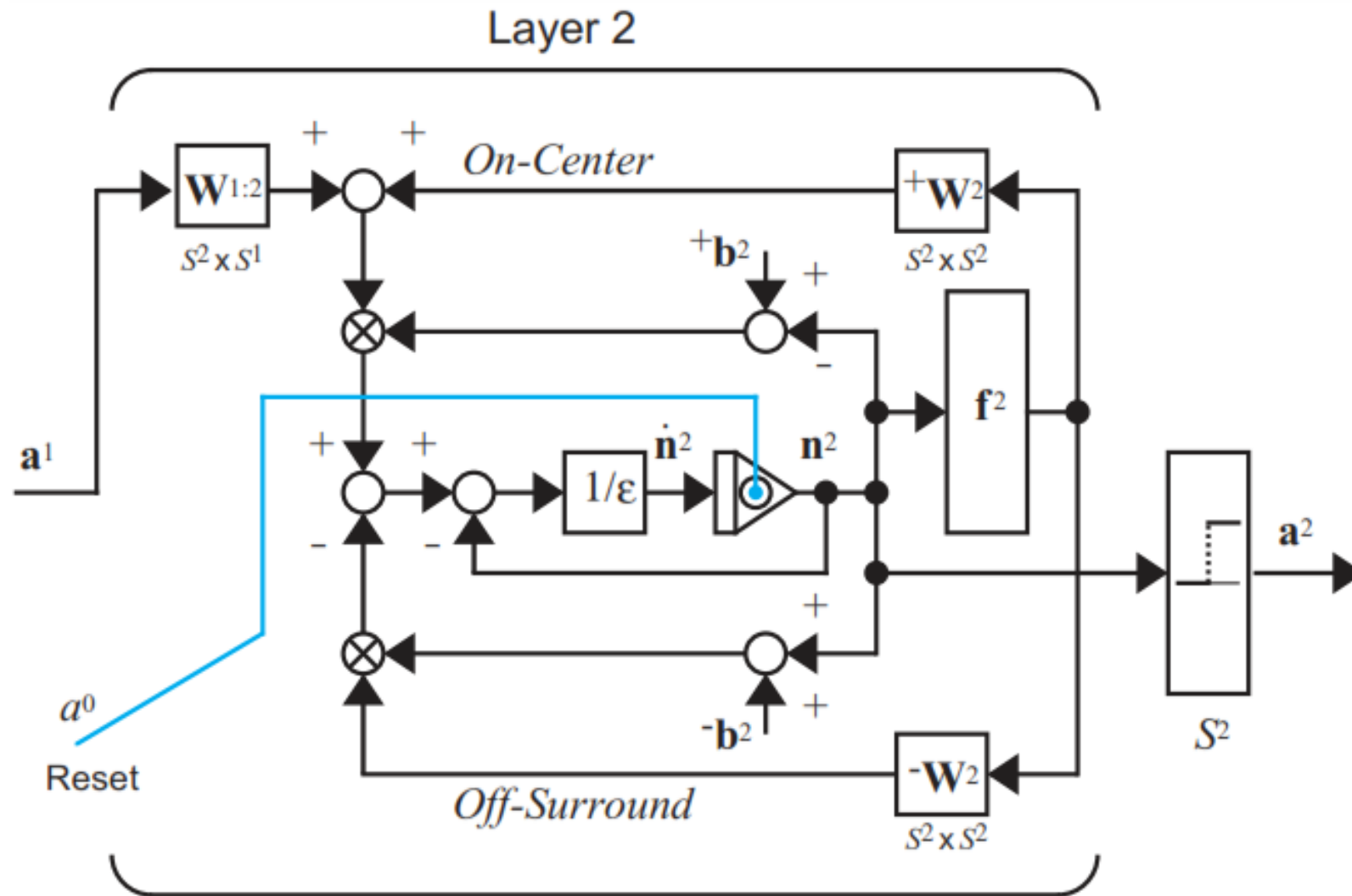
$$\mathbf{p} \cap \mathbf{w}_2^{2:1} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cap \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \mathbf{a}^1.$$



- Layer 2 of the ART1 is almost identical to Layer 2 of the Grossberg network.
- Its **main purpose is to contrast enhance its output pattern.**
- For our implementation of the ART1 network, the contrast enhancement will be a winner-take-all competition, so only the neuron that receives the largest input will have a nonzero output.
- There is one major difference between the second layers of the Grossberg and the ART1 networks.
 - Layer 2 of the ART1 network uses an integrator that can be **reset.**

- The reset signal, a^0 , is the output of the orienting subsystem.
- It generates a reset whenever there is a mismatch at Layer 1 between the input signal and the L2-L1 expectation.
- One other small difference between Layer 2 of ART1 and Layer 2 Grossberg network is that two transfer functions are used in ART1.
- The transfer function $f^2(n^2)$ is used for the on-center/off-surround feedback connections.
- The output of Layer 2 is computed as
$$a^2 = \text{hardlim}^+(n^2)$$
- The reason for the second transfer function is that we want the output of Layer 2 to be a binary signal.

Layer 2



Equation of operation of Layer 2

Shunting Model

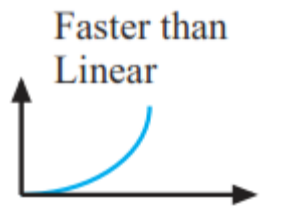
$$\begin{aligned} \varepsilon \frac{d\mathbf{n}^2(t)}{dt} = & -\mathbf{n}^2(t) \\ & + (\mathbf{b}^2 - \mathbf{n}^2(t)) \underbrace{\left\{ \overbrace{[\mathbf{W}^2]^+}^{\text{On-Center Feedback}} \mathbf{f}^2(\mathbf{n}^2(t)) + \overbrace{\mathbf{W}^{1:2} \mathbf{a}^1}^{\text{Adaptive Instars}} \right\}}_{\text{Excitatory input}} \\ & - (\mathbf{n}^2(t) + \mathbf{b}^2) \underbrace{[\mathbf{W}^2]^- \mathbf{f}^2(\mathbf{n}^2(t))}_{\substack{\text{Off-Surround} \\ \text{Feedback}}}_{\text{Inhibitory input}} \end{aligned}$$

The rows of $\mathbf{W}^{1:2}$, after training, will represent the prototype patterns.

Layer 2 Example

To illustrate the performance of Layer 2, consider a two-neuron layer with

$$\varepsilon = 0.1 \quad {}^+\mathbf{b}^2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad {}^-\mathbf{b}^2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{W}^{1:2} = \begin{bmatrix} ({}_1\mathbf{w}^{1:2})^T \\ ({}_2\mathbf{w}^{1:2})^T \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 1 & 0 \end{bmatrix}$$

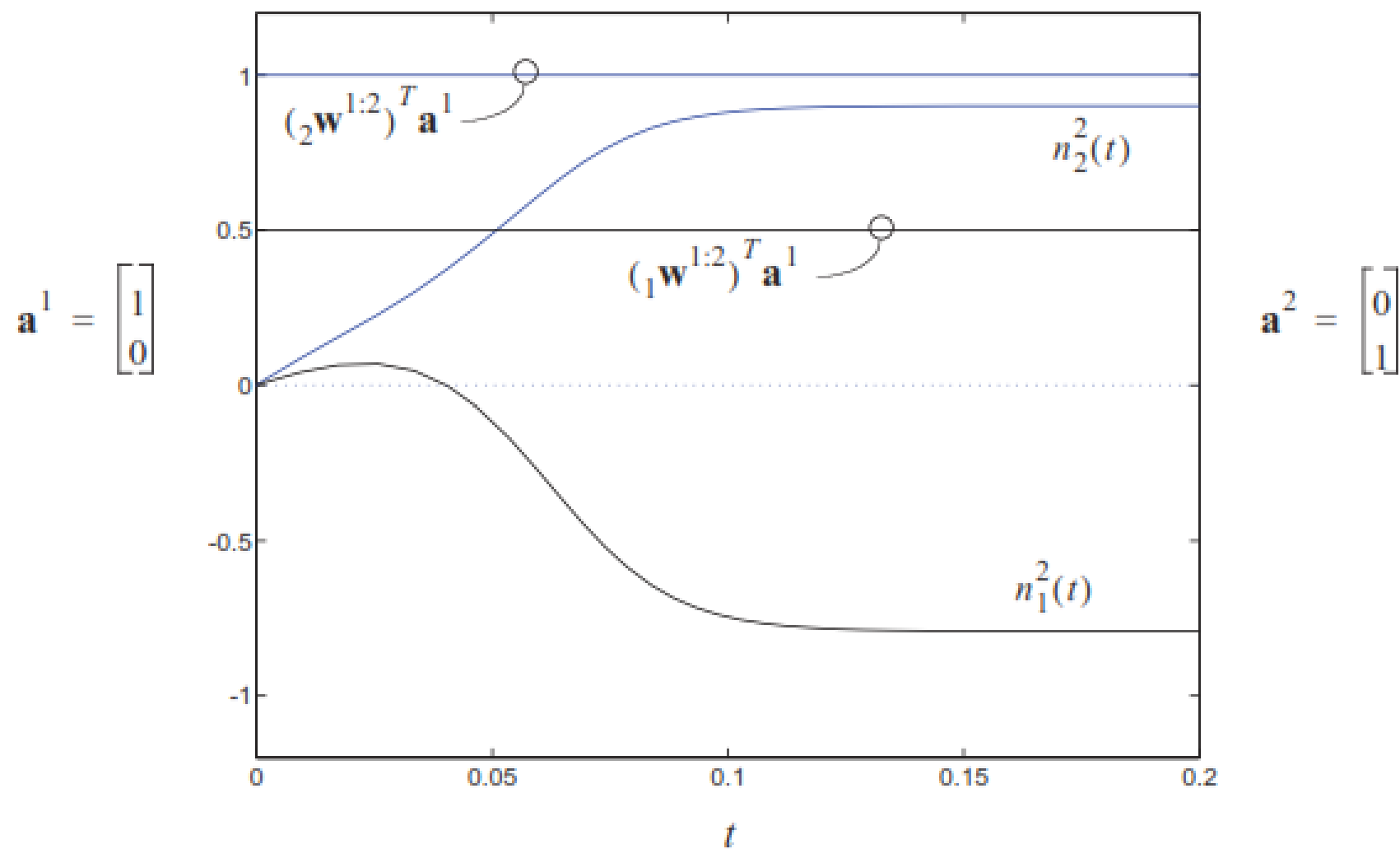


$$f^2(n) = \begin{cases} 10(n)^2, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad \text{(Faster than linear, winner-take-all)}$$

$$(0.1) \frac{dn_1^2(t)}{dt} = -n_1^2(t) + (1 - n_1^2(t)) \left\{ f^2(n_1^2(t)) + ({}_1\mathbf{w}^{1:2})^T \mathbf{a}^1 \right\} - (n_1^2(t) + 1) f^2(n_2^2(t))$$

$$(0.1) \frac{dn_2^2(t)}{dt} = -n_2^2(t) + (1 - n_2^2(t)) \left\{ f^2(n_2^2(t)) + ({}_2\mathbf{w}^{1:2})^T \mathbf{a}^1 \right\} - (n_2^2(t) + 1) f^2(n_1^2(t)) .$$

Layer 2 Response



Layer 2 Steady State operation

$$a_i^2 = \begin{cases} 1, & \text{if } ((\mathbf{w}^{1:2})^T \mathbf{a}^1 = \max[(\mathbf{w}^{1:2})^T \mathbf{a}^1]) \\ 0, & \text{otherwise} \end{cases}$$